Roll No.....

EVEN END SEMESTER EXAMINATION JUNE 2025

Name of the Program: **B. Tech. Petroleum Engineering**

Semester: **IV**

Name of the Course: **Applied Sedimentology**

Course Code: **TPE – 404**

Time: **3 Hours**

Maximum Marks: **100**

Note:

i) All questions are compulsory.
ii) Answer any two sub questions among **a, b & c** in each main question.
iii) Total marks for each main question are twenty.
iv) Each questions carries **10 Marks**

Q1		**(20 marks)**
(a)	Explain the concept of weathering and outline the main types of physical weathering.	CO1 CO2
(b)	What is meant by the processes of lithification and cementation, and how do they shape the reservoir properties—such as porosity and permeability—of sedimentary rocks?	
(c)	Differentiate between laminar and turbulent flows. How do they affect sediment transport?	
Q2		**(20 marks)**
(a)	Describe in detail about the agents involved in the transportation of sediments.	CO3 CO4
(b)	Analyze how the distance of sediment transport affects the grain size, sphericity, and roundness of sediment particles.	
(c)	Explain the following: a) Ripple marks. b) Reynolds's Numbers	
Q3		**(20 marks)**
(a)	Discuss the formation of graded bedding and its significance in interpreting sedimentary environments.	CO4 CO5
(b)	What are the different types of sedimentary structures, and how do they inform us about the sediment transport and deposition process?	
(c)	Discuss about Primary sedimentary structures. Also highlight their significance.	
Q4		**(20 marks**
(a)	Define depositional environment. Illustrate the physical and chemical parameters of depositional environments.	CO5
(b)	Outline the different sedimentary depositional environments and explain the features associated with continental environments.	
(c)	What is the significance of understanding sedimentary environments for resource exploration, such as petroleum or coal deposits?	
Q5		**(20 mark**
(a)	Describe a sedimentary basin and explain the different types of basins created by tectonic actions.	CO6
(b)	Describe the processes of mechanical stretching, thermal subsidence, and flexural loading, and their contributions to basin formation.	
(c)	Define sedimentary basin. Discuss the categories of Indian sedimentary basins.	